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### Light radiating device

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#### Abstract

A light radiating device (1A) performs solidification or cauterization of a biological tissue (PL) by radiating a light beam (BM). A light source (10A) emits the light beam (BM). An optical waveguide (20A) is a member being provided with a reflection surface (21A) totally reflecting the light beam (BM) on an inner circumferential side wall, causing the light beam (BM) emitted from the light source (10A) to enter a part enclosed by the inner circumferential side wall from one end, and sending the light beam (BM) to the other end. A catoptric system (30A) reflects the light beam (BM) sent to the other end of the optical waveguide (20A) and condenses the light beam (BM) on the biological tissue (PL).

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| <b>Family ID:</b>            | <b>69163744</b>                                                                                                                  |
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## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name   | U.S. Cl. | CPC            |
|--------------|-------------|-----------------|----------|----------------|
| 5370643      | 12/1993     | Krivoshlykov    | 385/124  | G02B<br>6/4296 |
| 5437660      | 12/1994     | Johnson         | 606/7    | A61B<br>18/24  |
| 5836941      | 12/1997     | Yoshihara       | 606/17   | A61B<br>18/24  |
| 6168590      | 12/2000     | Neev            | 606/9    | A61B<br>18/203 |
| 6379347      | 12/2001     | Maki            | 606/17   | A61B<br>18/24  |
| 6607526      | 12/2002     | Maki            | 600/101  | A61N<br>5/0601 |
| 2002/0045811 | 12/2001     | Kittrell        | 606/7    | G02B<br>6/4296 |
| 2006/0084958 | 12/2005     | Raif            | 606/17   | A61B<br>18/148 |
| 2007/0282315 | 12/2006     | Strassl         | 606/9    | A61C<br>1/052  |
| 2008/0308753 | 12/2007     | Stuba et al.    | N/A      | N/A            |
| 2011/0087202 | 12/2010     | Lewinsky        | 606/14   | A61N<br>5/0603 |
| 2013/0274726 | 12/2012     | Takayama        | 606/15   | G02B<br>6/262  |
| 2015/0283397 | 12/2014     | Andersen et al. | N/A      | N/A            |
| 2018/0049806 | 12/2017     | Yu              | N/A      | A61B<br>18/22  |

## FOREIGN PATENT DOCUMENTS

| Patent No.    | Application Date | Country | CPC        |
|---------------|------------------|---------|------------|
| S6211820      | 12/1986          | JP      | N/A        |
| H04285548     | 12/1991          | JP      | N/A        |
| H0838499      | 12/1995          | JP      | N/A        |
| H0951899      | 12/1996          | JP      | N/A        |
| 2000-225123   | 12/1999          | JP      | N/A        |
| 2005-079197   | 12/2004          | JP      | N/A        |
| 2005-177150   | 12/2004          | JP      | N/A        |
| 2010-529885   | 12/2009          | JP      | N/A        |
| 2017-515530   | 12/2016          | JP      | N/A        |
| WO-2016148718 | 12/2015          | WO      | A61B 18/22 |

## OTHER PUBLICATIONS

“Ellipsoid” Britannica (Year: 2006). cited by examiner

“Paraboloid” Britannica (Year: 2008). cited by examiner

International Search Report (International Application No. PCT/JP2019/028471); Mail Date: Oct. 8, 2019; 4 pages; Includes English Translation. cited by applicant

Written Opinion (International Application No. PCT/JP2019/028471); Mail Date Oct. 8, 2019; 5 pages. cited by applicant

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a national phase application of International Patent Application No. PCT/JP2019/028471, filed Jul. 19, 2019, which claims priority to JP Patent Application No. 2018-137030, filed Jul. 20, 2018, the disclosures of which are hereby incorporated by reference in their entirety.

### TECHNICAL FIELD

(2) The present disclosure relates to a light radiating device.

### BACKGROUND ART

(3) A device enabling solidification or cauterization of a biological tissue is disclosed in Patent Literature 1. In the device, light being output from a halogen light source and propagating through an optical fiber member is output from a metal sleeve provided at one end of the optical fiber member and is radiated to a biological tissue. Resulting generation of heat in the biological tissue enables solidification or cauterization of the biological tissue.

### CITATION LIST

#### Patent Literature

(4) Patent Literature 1: Unexamined Japanese Patent Application Publication (Translation of PCT Application) No. 2010-529885

### SUMMARY OF INVENTION

#### Technical Problem

(5) In the aforementioned device, light output from the metal sleeve is reflected off an inner wall of

the optical fiber member and is diffused. Accordingly, heat is generated by the light only around the metal sleeve. Therefore, in order to perform solidification or cauterization of a biological tissue by use of the aforementioned device, the metal sleeve needs to be in tight contact with the biological tissue (radiation target).

(6) The present disclosure has been made in view of the circumstance described above, and the objective thereof is to provide a light radiating device that can perform solidification or cauterization of a radiation target in a state of being kept apart from the radiation target without being in tight contact with the radiation target.

#### Solution to Problem

(7) In order to achieve the aforementioned objective, a light radiating device according to the present disclosure is a light radiating device performing solidification or cauterization of a radiation target by radiating a light beam, the light radiating device including: a light source emitting the light beam; an optical waveguide having an inner circumferential side wall provided with a reflection surface totally reflecting the light beam, causing the light beam emitted from the light source to enter a part enclosed by the inner circumferential side wall from one end, and sending the light beam to another end; and a catoptric system reflecting the light beam sent to another end of the optical waveguide and condensing the light beam on the radiation target.

(8) In this case, a reflection surface of the catoptric system may condense the reflected light beam with respect to a direction along an incidence plane of a light ray entering from a direction of an optical axis of the optical waveguide.

(9) Further, a sectional shape of a reflection surface of the catoptric system when the reflection surface is cut by an incidence plane of a light ray entering along an optical axis of the optical waveguide may be a concave elliptical arc, and a reflection surface of the catoptric system may condense a light ray entering each point on the elliptical arc from a direction of an optical axis of the optical waveguide on a focus of the elliptical arc.

(10) A sectional shape of a reflection surface of the catoptric system when the reflection surface is cut by a plane including an incidence position of a light ray entering along an optical axis of the optical waveguide and being orthogonal to an optical axis of the optical waveguide may be a concave circular arc centered on the focus, and a reflection surface of the catoptric system may condense a light ray entering each point on the circular arc from a direction parallel to the optical axis on the focus.

(11) A sectional shape of a reflection surface of the catoptric system when the reflection surface is cut by a plane including an incidence position of a light ray entering along an optical axis of the optical waveguide and being orthogonal to an optical axis of the optical waveguide may be a straight line or a convex curve, and a reflection surface of the catoptric system may reflect, toward a linear region including the focus on the radiation target, a light ray entering each point on the straight line or the convex curve from a direction parallel to the optical axis.

(12) A reflection surface of the catoptric system may be formed by a plurality of planes normal line directions of which are different from one another being connected.

(13) A section of a reflection surface of the catoptric system when the reflection surface is cut by a plane orthogonal to an optical axis of the optical waveguide may be a polygonal line.

(14) A section of a reflection surface of the catoptric system when the reflection surface is cut by an incidence plane of a light ray entering along an optical axis of the optical waveguide may be a polygonal line.

(15) When a light ray an emission angle of which from the light source with respect to an optical axis of the optical waveguide is maximum is reflected at a second position on the inner circumferential side wall after being reflected at a first position on the inner circumferential side wall, a length of the optical waveguide in a direction of an optical axis may be an integral multiple of a distance between the first position and the second position with respect to the optical axis.

(16) A reflection surface of the optical waveguide may be a surface of a metal material film formed

on the inner circumferential side wall, and a film made up of silicon dioxide may be formed on a surface of the metal material film.

(17) A thickness of the film made up of silicon dioxide may be  $\frac{1}{4}$  of a wavelength of a part of one or more light rays included in a light beam emitted from the light source.

(18) A first optical condensing system condensing the light beam emitted from the light source and causing the light beam to enter at one end of the optical waveguide may be provided.

(19) A second optical condensing system condensing the light beam emitted from another end of the optical waveguide and causing the light beam to enter the catoptric system may be provided.

(20) A distance from the second optical condensing system to the catoptric system may be greater than a focal distance of the second optical condensing system.

(21) When a sectional shape of a concave reflection surface of the catoptric system when the concave reflection surface is cut by an incidence plane of a light ray entering along an optical axis of the optical waveguide is approximated to a parabola, a focus position of the second optical condensing system may be positioned at a focus of the parabola.

(22) The inner circumferential side wall may be formed in a prismatic shape in the optical waveguide.

(23) The optical waveguide may be a hollow tube.

(24) The optical waveguide may be a quartz rod.

(25) A heat insulating part may be provided on an outer periphery of the optical waveguide and the catoptric system.

(26) The heat insulating part may include: an exterior cover enclosing an outer periphery of the optical waveguide and the catoptric system; and a spacer inserted between the optical waveguide and the catoptric system, and the exterior cover in order to provide a gap between the optical waveguide and the catoptric system, and the exterior cover.

(27) On the spacer, a plurality of projections may be provided on at least one of a part facing the optical waveguide and the catoptric system, and a part facing the exterior cover.

(28) A size of a reflection surface of the catoptric system when the reflection surface is viewed from a direction of an optical axis of the optical waveguide may be equal to or greater than a size of a section of a light beam entering the catoptric system from the optical waveguide.

(29) A light beam emitted from the light source may include an infrared radiation.

(30) The light source may be one of a halogen lamp, a xenon lamp, an infrared heater, and an infrared laser.

#### Advantageous Effects of Invention

(31) The present disclosure includes a catoptric system reflecting a light beam sent to another end of an optical waveguide and condensing the light beam. The light beam can be thereby condensed at a position apart from the device and can generate intense heat at the position. Consequently, solidification or cauterization of a radiation target can be performed in a state of being kept apart from the radiation target without being in tight contact with the radiation target.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a top view illustrating a structure of a light radiating device according to Embodiment 1 of the present disclosure;

(2) FIG. 2 is a cross-sectional view taken along an A-A line in FIG. 1;

(3) FIG. 3 is an expanded sectional view of a catoptric system along the optical axis of an optical waveguide;

(4) FIG. 4 is a schematic diagram of an ellipse;

(5) FIG. 5 is a cross-sectional view taken along a B-B line in FIG. 1;

(6) FIG. 6 is a schematic diagram illustrating the three-dimensional shape of a reflection surface;

(7) FIG. 7 is a diagram illustrating the length of the optical waveguide in the optical axis direction;

(8) FIG. 8A is a perspective view illustrating a partial structure of a light radiating device according to Embodiment 2 of the present disclosure;

(9) FIG. 8B is a front view of the light radiating device in FIG. 8A;

(10) FIG. 8C is a side view illustrating a partial structure of the light radiating device in FIG. 8A;

(11) FIG. 9 is a cross-sectional view taken along a C-C line in FIG. 8B and a cross-sectional view taken along a D-D line in FIG. 8C;

(12) FIG. 10 is a schematic diagram illustrating the three-dimensional shape of a reflection surface;

(13) FIG. 11 is a schematic diagram illustrating a case of performing solidification or cauterization of a tubular biological tissue;

(14) FIG. 12A is a first diagram illustrating a structure of an optical waveguide constituting a light radiating device according to Embodiment 3 of the present disclosure;

(15) FIG. 12B is a second diagram illustrating the structure of the optical waveguide constituting the light radiating device according to Embodiment 3 of the present disclosure;

(16) FIG. 12C is a third diagram illustrating the structure of the optical waveguide constituting the light radiating device according to Embodiment 3 of the present disclosure;

(17) FIG. 13A is a first diagram illustrating a structure of an optical waveguide the sectional shape of which is a circle;

(18) FIG. 13B is a second diagram illustrating the structure of the optical waveguide the sectional shape of which is a circle;

(19) FIG. 13C is a cross-sectional view taken along a C-C line in FIG. 13A;

(20) FIG. 14 is a perspective view illustrating a partial structure of a light radiating device according to Embodiment 4 of the present disclosure;

(21) FIG. 15A is a top view illustrating a structure of a light radiating device according to Embodiment 5 of the present disclosure;

(22) FIG. 15B is a cross-sectional view taken along an E-E line in FIG. 15A;

(23) FIG. 16 is a cross-sectional view illustrating a shape of a spacer;

(24) FIG. 17 is a cross-sectional view illustrating a modified example (1) of the shape of the spacer;

(25) FIG. 18 is a cross-sectional view illustrating a modified example (2) of the shape of the spacer;

(26) FIG. 19 is a perspective view illustrating a structure of a light radiating device according to Embodiment 6 of the present disclosure;

(27) FIG. 20 is a perspective view illustrating a structure of a light radiating device according to Embodiment 7 of the present disclosure;

(28) FIG. 21 is a diagram of a catoptric system in FIG. 20 viewed from the +z-direction;

(29) FIG. 22 is a cross-sectional view taken along an F-F line in FIG. 21;

(30) FIG. 23 is a graph illustrating changes in a cauterization depth with respect to cauterization time;

(31) FIG. 24 is a graph illustrating changes in a cauterization volume with respect to cauterization time;

(32) FIG. 25 is a perspective view illustrating a shape (1) of a reflection surface of a catoptric system;

(33) FIG. 26 is a perspective view illustrating a shape (2) of the reflection surface of the catoptric system;

(34) FIG. 27 is a perspective view illustrating a shape (3) of the reflection surface of the catoptric system;

(35) FIG. 28 is a diagram illustrating the structure of the light radiating device according to Embodiment 7 of the present disclosure;

(36) FIG. 29 is a diagram illustrating a structure around a first optical condensing system;

(37) FIG. 30A is a diagram illustrating a structure around a second optical condensing system;

- (38) FIG. 30B is a diagram illustrating a modified example of the structure around the second optical condensing system;
- (39) FIG. 31 is a diagram illustrating a modified example of the structure of the light radiating device according to Embodiment 7 of the present disclosure;
- (40) FIG. 32 is a diagram illustrating a structure of the light radiating device in FIG. 28 when a catoptric system is removed;
- (41) FIG. 33 is a schematic diagram illustrating a relation between the catoptric system and a light beam BM emitted from an optical waveguide;
- (42) FIG. 34A is a cross-sectional view illustrating a modified example of a reflection surface; and
- (43) FIG. 34B is an expanded sectional view of a part E of the reflection surface in FIG. 34A.

#### DESCRIPTION OF EMBODIMENTS

(44) Light radiating devices according to embodiments of the present disclosure will be described in detail below with reference to drawings. In every drawing, the same or equivalent components are given the same sign. The light radiating device according to the present embodiment can be considered as an optical scalpel performing solidification or cauterization of a biological tissue in a state of separating the biological tissue being a radiation target from the device.

##### Embodiment 1

- (45) First, Embodiment 1 of the present disclosure will be described. As illustrated in FIG. 1, a light radiating device 1A according to Embodiment 1 is a long and narrow columnar member. As illustrated in FIG. 2, the light radiating device 1A includes a light source 10A, an optical waveguide 20A, and a catoptric system 30A.
- (46) The light source 10A is a surface light source and emits a light beam BM being a beam of light rays IL from a part facing one end of the optical waveguide 20A. For example, an infrared laser is used as the light source 10A. Accordingly, a wavelength region of the emitted light beam BM mainly includes the infrared region and more specifically the near-infrared region.
- (47) The optical waveguide 20A is an optical member extending in a direction of an optical axis AX1 thereof. The optical waveguide 20A causes the light beam BM emitted from the light source 10A to enter from one end and emits the light beam BM from the other end.
- (48) The optical waveguide 20A according to the present embodiment is a tubular (hollow) member and has an inner circumferential side wall. The sectional shape of the optical waveguide 20A is a circle. A reflection surface 21A totally reflecting the light beam BM is provided on the inner circumferential side wall of the optical waveguide 20A. Part of the light beam BM emitted from the light source 10A enters a part enclosed by the inner circumferential side wall (reflection surface 21A) and is sent to the other end while being totally reflected off the reflection surface 21A. An enclosure of the optical waveguide 20A is formed of metal or resin, and the reflection surface 21A is a surface of a film formed by depositing metal such as aluminum on the inner circumferential side wall.
- (49) The catoptric system 30A reflects the light beam BM sent to the other end of the optical waveguide 20A and condenses the light beam BM on a focus F on an optical axis AX2. Specifically, a reflection surface 31A of the catoptric system 30A condenses the reflected light beam BM with respect to an x-axis direction along an incidence plane of a light ray IL entering from a direction of the optical axis AX1 of the optical waveguide 20A. An enclosure of the catoptric system 30A is formed of metal or resin, and the reflection surface 31A is formed by depositing metal such as aluminum.
- (50) In the catoptric system 30A, the sectional shape of the reflection surface 31A when the reflection surface is cut by an incidence plane IP1 of a light ray entering along the optical axis AX1 of the optical waveguide 20A (a sectional shape along the x-axis direction) is a concave elliptical arc 32, as illustrated in FIG. 3. The elliptical arc 32 is part of an ellipse as illustrated in FIG. 4. In the ellipse in FIG. 4, the length of the major axis along an x'-axis direction is denoted by 2a, and the length of the minor axis along a y'-axis direction is denoted by 2b. The foci of the ellipse are

denoted by F.sub.1 and F.sub.2. At this time, light parallel to the x'-axis direction is condensed on the focus F.sub.1 or F.sub.2 of the ellipse.

(51) The reflection surface **31A** condenses a light ray IL entering each point on the elliptical arc **32** from the direction of the optical axis AX1 of the optical waveguide **20A** on a focus F of the elliptical arc **32**. The focus F matches the focus F.sub.1 or F.sub.2 of the ellipse illustrated in FIG. 4.

(52) Further, in the catoptric system **30A**, the sectional shape of the reflection surface **31A** when the reflection surface **31A** is cut by a plane IP2 including an incidence position P of a light ray IL entering along the optical axis AX1 of the optical waveguide **20A** and being orthogonal to the optical axis AX1 of the optical waveguide **20A** is a concave circular arc **33** centered on the focus F, as illustrated in FIG. 5. The reflection surface **31A** condenses a light ray IL entering each point on the circular arc **33** from a direction parallel to the optical axis AX1 on the focus F. A radius R of the circular arc **33** is determined in such a way that a light ray IL is condensed on the focus F.

(53) As described above, the sectional shape of the reflection surface **31A** of the catoptric system **30A** along the x-axis direction is the elliptical arc **32**, and the sectional shape along a y-axis direction is the circular arc **33**. The three-dimensional shape of the reflection surface **31A** having such sectional shapes is as illustrated in FIG. 6. The three-dimensional shape of the reflection surface **31A** is a curved surface formed when each point on the elliptical arc **32** is rotated around the focus F. Every light ray IL entering in the x-axis direction is thereby condensed on the focus F. Note that the outline of the reflection surface **31A** is assumed to be a rectangle in FIG. 6.

(54) Thus, in the light radiating device **1A** according to the present embodiment, out of light rays IL constituting the light beam BM emitted from the light source **10A**, a light ray IL traveling in parallel with the optical axis AX1 is condensed on the focus F.

(55) In practice, light rays IL emitted from the light source **10A** do not necessarily travel along the optical axis AX1 and travel in such a way as to diffuse from each point of the light source **10A**. As illustrated in FIG. 7, the maximum emission angle of a light ray IL with respect to the optical axis AX1 of the optical waveguide **20A** is denoted by  $\theta$ . For example,  $\theta$  is 30 degrees.

(56) It is assumed here that a light ray IL the emission angle of which from the light source **10A** with respect to the optical axis AX1 of the optical waveguide **20A** is the maximum emission angle  $\theta$  is reflected at a second position P2 on the reflection surface **21A** after being reflected at a first position P1 on the reflection surface **21A**, as illustrated in FIG. 7. In this case, it is assumed in the present embodiment that the length of the optical waveguide **20A** in the optical axis AX1 direction is a length L being four times a distance a between the first position P1 and the second position P2 with respect to the optical axis AX1.

(57) A light ray IL emitted from the light source **10A** in a diffusing manner enters, in the x-axis direction, a region AR centered on the focus F on a biological tissue PL being a radiation target. Since the length L of the optical waveguide **20A** in the optical axis AX1 direction is an integral multiple of the distance a, intensity of the light ray IL entering the region AR is maximum at the focus F and gradually decreases as the distance from the focus F increases. An intensity distribution of the light ray IL on the biological tissue PL is rotationally symmetric around the focus F.

(58) Thus, the light radiating device **1A** according to the present embodiment condenses the light beam BM on the focus F in the x-axis direction and the y-axis direction. Light intensity on the focus F is thereby increased, and a heating value increases. Consequently, solidification or cauterization of the biological tissue PL being the radiation target at the focus F can be performed.

## Embodiment 2

(59) Next, Embodiment 2 of the present disclosure will be described. A light radiating device **1B** according to the present embodiment differs from the light radiating device **1A** according to aforementioned Embodiment 1 in including a catoptric system **30B** in place of the catoptric system **30A**, as illustrated in FIG. 8A, FIG. 8B, and FIG. 8C.

(60) As illustrated in FIG. 9, a characteristic that a reflection surface **31B** of the catoptric system



**30B** condenses a reflected light beam **BM** with respect to an x-axis direction along an incidence plane of a light ray **IL** entering from a direction of an optical axis **AX1** of an optical waveguide **20A** in the light radiating device **1B** according to the present embodiment is the same as that of the catoptric system **30A** according to aforementioned Embodiment 1.

(61) Specifically, in the catoptric system **30B**, the sectional shape of the reflection surface **31B** when the reflection surface **31B** is cut by an incidence plane **IP1** of a light ray entering along the optical axis **AX1** of the optical waveguide **20A** is a concave elliptical arc **32**, as illustrated in FIG. **9**, similarly to the catoptric system **30A**.

(62) However, in the catoptric system **30B** according to the present embodiment, the sectional shape of the reflection surface **31B** when the reflection surface **31B** is cut by a plane **IP2** including an incidence position **P** of a light ray **IL** entering along the optical axis **AX1** of the optical waveguide **20A** and being orthogonal to the optical axis **AX1** of the optical waveguide **20A** is a convex curve (circular arc **34**), as illustrated in FIG. **9**. The reflection surface **31B** reflects, toward a linear region **AR** including a focus **F** on a biological tissue **PL** being a radiation target, a light ray **IL** entering each point on the convex curve (circular arc **34**) from a direction parallel to the optical axis **AX1** of the optical waveguide **20A**.

(63) The sectional shape of the reflection surface **31B** along a y-axis direction orthogonal to the x-axis direction may be a straight line, according to the present embodiment. Even in such a case, the sectional shape of the light beam **BM** on the biological tissue **PL** being the radiation target is the linear region **AR**.

(64) As described above, the sectional shape of the reflection surface **31B** of the catoptric system **30B** along the x-axis direction is the elliptical arc **32**, and the sectional shape along the y-axis direction is the circular arc **34**. The three-dimensional shape of the reflection surface **31B** having such sectional shapes is as illustrated in FIG. **10**. A curve **CL** (a curve similar to the elliptical arc **32**) projecting the elliptical arc **32** viewed from the center (focus **F**) of the linear region **AR** on the biological tissue **PL** is defined as illustrated in FIG. **10**. Then, a curved surface formed when rotating each point on the elliptical arc **32** around the intersection of a straight line connecting each point on the elliptical arc **32** and the focus **F**, and the curve **CL** is the reflection surface **31B**. Note that the outline of the reflection surface **31B** is assumed to be a rectangle in FIG. **10**.

(65) Thus, the light radiating device **1B** according to the present embodiment condenses the light beam **BM** on the linear region **AR**. Light intensity on the region **AR** is thereby increased, and a heating value increases. Consequently, solidification or cauterization of the biological tissue **PL** being a radiation target can be performed.

(66) For example, the light radiating device **1B** according to the present embodiment is suitable for performing solidification or cauterization of an inner circumferential wall of a tubular biological tissue **PL** along the circumference, as illustrated in FIG. **11**.

### Embodiment 3

(67) Next, Embodiment 3 of the present disclosure will be described. A light radiating device **1C** according to the present embodiment differs from the light radiating devices **1A** and **1B** according to the aforementioned embodiments in using an optical waveguide **20B** in place of the optical waveguide **20A**, as illustrated in FIG. **12A**, FIG. **12B**, and FIG. **12C**.

(68) An inner circumferential side wall of the optical waveguide **20B** is formed in a prismatic shape. For example, the section of the inner circumferential side wall is a hexagon, as illustrated in FIG. **12B**.

(69) FIG. **13A**, FIG. **13B**, and FIG. **13C** illustrate an optical waveguide **20C** having the same diameter as the optical waveguide **20B**, the section of the optical waveguide **20C** being a circle. As comparison between a light intensity distribution in FIG. **12C** and a light intensity distribution in FIG. **13C** tells, forming the optical waveguide **20B** in a prismatic shape allows a light intensity distribution of the output light beam **BM** to be more uniform compared with the optical waveguide **20C** the section of which is a circle.

(70) The light radiating device **1C** according to the present embodiment may include the catoptric system **30A** or may include the catoptric system **30B**.

#### Embodiment 4

(71) Next, Embodiment 4 of the present disclosure will be described. A light radiating device **1D** according to the present embodiment differs from the light radiating device **1A** according to aforementioned Embodiment 1 in using an optical waveguide **20D** in place of the optical waveguide **20A**, as illustrated in FIG. **14**. Furthermore, the optical waveguide **20D** according to the present embodiment is not directly connected to a light source **10A** and is connected through an optical fiber **40**.

(72) While each of the optical waveguides **20A**, **20B**, and **20C** according to the aforementioned embodiments is a tubular optical member, the optical waveguide **20D** according to the present embodiment is a quartz rod, as illustrated in FIG. **14**. The quartz rod also allows causing a light beam **BM** emitted from the light source **10A** to enter from one end through the optical fiber **40**, totally reflecting the light beam **BM** off an inner circumferential side wall, and emitting the light beam **BM** from the other end. Note that illustration of a catoptric system **30A** is omitted in FIG. **14**.

(73) The refractive index and the width of the optical waveguide **20D** needs to be set in such a way that a light ray **IL** satisfies a total reflection condition with respect to the maximum emission angle  $\theta$  from the light source **10A**.

(74) While the sectional shape of the optical waveguide **20D** is a hexagon, the sectional shape may be a triangle, a rectangle, or a circle.

(75) However, intensity of a light ray **IL** emitted from the light source **10A** attenuates more in the quartz rod compared with a hollow tubular member. Therefore, when an intensive light beam **BM** is required, it is desirable to use the hollow tubular optical waveguide **20A** or **20B**.

(76) The light radiating device **1D** according to the present embodiment may include the catoptric system **30A** or may include the catoptric system **30B**.

#### Embodiment 5

(77) Next, Embodiment 5 of the present disclosure will be described. As illustrated in FIG. **15B** being a cross-sectional view taken along an E-E line in FIG. **15A**, a light radiating device **1E** according to the present embodiment differs from each of the aforementioned embodiments in including a heat insulating part **50**.

(78) The heat insulating part **50** is provided on an outer periphery of an optical waveguide **20A** and a catoptric system **30A**. The heat insulating part **50** includes an exterior cover **51** and a spacer **52**.

(79) The exterior cover **51** encloses the outer periphery of the optical waveguide **20A** and the catoptric system **30A**. The spacer **52** is inserted between the optical waveguide **20A** and the catoptric system **30A**, and the exterior cover **51** in order to provide a gap between the optical waveguide **20A** and the catoptric system **30A**, and the exterior cover **51**.

(80) Heat is generated in the optical waveguide **20A** and the catoptric system **30A** when a light beam **BM** is reflected. It is undesirable that the heat be transferred to a surrounding biological tissue not being a radiation target. Then, the light radiating device **1E** according to the present embodiment provides heat insulation by an air layer formed by the exterior cover **51** and the spacer **52** and suppresses heat transfer to the surrounding biological tissue.

(81) As illustrated in FIG. **16**, the spacer **52** is a C-shaped member and allows heat insulating air between the catoptric system **30A** and the exterior cover **51** to move inside. Therefore, the heat insulation effect can be enhanced by recovery and supply of the heat insulating air, that is, air circulation.

(82) Various shapes may be considered as the shape of the spacer **52**. For example, a plurality of projections **53** may be provided on a part facing the optical waveguide **20A** and the catoptric system **30A**, and a part facing the exterior cover **51** on the spacer **52**, as illustrated in FIG. **17**. Thus, a contact area between the spacer **52**, and the optical waveguide **20A** and the catoptric system **30A**, and a contact area between the spacer **52** and the exterior cover **51** can be reduced, and the

heat insulation effect can be enhanced. In this case, the projections **53** may be provided on at least one of the part facing the optical waveguide **20A** and the catoptric system **30A**, and the part facing the exterior cover **51**.

(83) Further, the part facing the optical waveguide **20A** and the catoptric system **30A**, and the part facing the exterior cover **51** on the spacer **52** may be narrowed, as illustrated in FIG. **18**. Such narrowing can also reduce the contact area between the spacer **52**, and the optical waveguide **20A** and the catoptric system **30A**, and the contact area between the spacer **52** and the exterior cover **51**, and enhance the heat insulation effect.

(84) Further, a through hole **54** may be provided in the spacer **52**, as illustrated in FIG. **18**. A path of the heat insulating air can be formed by the through hole **54**.

(85) It is desirable that the exterior cover **51** and the spacer **52** be formed of a material with low heat conductivity. For example, biocompatible materials such as titan may be used as a material of the exterior cover **51** and the spacer **52**.

(86) Heat insulation by an air layer is provided in the present embodiment. In this case, as described above, heat insulation may be provided by circulating air in the air layer. Further, a refrigerant may be inserted in the gap, and the refrigerant may be circulated.

(87) The light radiating device **1E** according to the present embodiment may include the catoptric system **30B**.

#### Embodiment 6

(88) Next, Embodiment 6 of the present disclosure will be described. A light radiating device **1F** according to the present embodiment differs from aforementioned Embodiment 1 in a light source **10B** not being directly connected to an optical waveguide **20A**, as illustrated in FIG. **19**.

(89) The light radiating device **1F** according to the present embodiment includes the light source **10B** and a rod-shaped operation device **60**. The light source **10B** is a halogen lamp, a xenon lamp, or an infrared heater. A shutter **65** is mounted on the light source **10B**, and rotation of the shutter **65** allows a pulse-shaped light beam BM to be output from the light source **10B**.

(90) The optical waveguide **20A** and a catoptric system **30B** are provided inside the operation device **60**. The light source **10B** is connected to the operation device **60** by an optical fiber **80**. A light beam emitted from the light source **10B** is sent to the optical waveguide **20A** through the optical fiber **80**, is reflected by the catoptric system **30B**, and is radiated to a biological tissue PL being a radiation target.

(91) The operation device **60** is provided with a grip **61** and a switch **62**. An operator holds the grip **61** and operates the light radiating device **1F**. The switch **62** is a switch for turning on and off output of the light beam BM radiated from the light radiating device **1F**. The optical waveguide **20A** and the catoptric system **30B** are extendable and rotatable with respect to the operation device **60**. A position of a focus F or a region AR can be thereby adjusted according to a part being a radiation target.

(92) After the adjustment, the operator holds the grip **61**, turns on the switch **62** while adjusting the focus F of the catoptric system **30B** to a part being a radiation target, and radiates the light beam BM to the part.

(93) The light radiating device **1F** according to the present embodiment may include the optical waveguide **20B**, **20C**, or **20D** in place of the optical waveguide **20A**. Further, the light radiating device **1F** may include the catoptric system **30A** in place of the catoptric system **30B**. Further, the light radiating device **1F** may include the heat insulating part **50**.

#### Embodiment 7

(94) Next, Embodiment 7 of the present disclosure will be described. In each of the light radiating devices **1A** to **1F** according to the aforementioned embodiments, the shape of the reflection surface of the catoptric system **30A** or **30B** is a curved surface. Such a reflection surface requires highly skilled manufacturing and also requires highly skilled evaluation of the reflection surface after manufacturing. Then, the present embodiment provides a light radiating device including a

reflection surface not requiring highly skilled manufacturing and highly skilled evaluation.

(95) A light radiating device **1G** according to the present embodiment includes an optical waveguide **20E** and a catoptric system **30C**, as illustrated in FIG. **20**. The optical waveguide **20E** is a tubular member the section of which is a hexagon, but without being limited thereto, the optical waveguide according to each of the aforementioned embodiments may be used. The light source according to each of the aforementioned embodiments may also be used.

(96) The shape of a reflection surface of the catoptric system **30C** in the light radiating device **1G** according to the present embodiment differs from that according to each of the aforementioned embodiments. The reflection surface of the catoptric system **30C** is formed by connecting a plurality of planes the normal line directions of which are different from one another.

(97) As illustrated in FIG. **21**, the catoptric system **30C** includes a plane mirror **30C1** and a plane mirror **30C2**. As illustrated in FIG. **22**, each of a reflection surface **31C1** of the plane mirror **30C1** and a reflection surface **31C2** of the plane mirror **30C2** is a plane. The reflection surface **31C1** and the reflection surface **31C2** are connected by a connecting line DL parallel to an x-z plane in a state of a normal line direction ND1 and a normal line direction ND2 of the respective reflection surfaces being different from each other and more specifically in a state of an angle formed between the reflection surface **31C1** and the reflection surface **31C2** being less than 180°. The sectional shape of a reflection surface **31C** when the reflection surface **31C** is cut by a y-z plane orthogonal to an optical axis AX1 of the optical waveguide **20E** thereby becomes a concave polygonal line shape. The catoptric system **30C** condenses an entering light ray with respect to a light beam in a y-axis direction.

(98) A cauterization characteristic of a biological tissue PL will be compared between the light radiating device **1G** using the catoptric system **30C** (a polygonal plane mirror) according to the present embodiment and a light radiating device using a concave surface approximated to the reflection surface **31C** of the light radiating device **1G** as a reflection surface. The length of the optical waveguide **20E** is assumed to be the same (such as 100 mm) between the two devices.

(99) As illustrated in FIG. **23**, changes in a cauterization depth of the biological tissue PL change parabolically with respect to cauterization time. The rate of change is almost the same between the case of using the polygonal plane mirror and the case of using the concave surface.

(100) Further, as illustrated in FIG. **24**, changes in a cauterization volume of the biological tissue PL also change parabolically with respect to cauterization time in the case of using the polygonal plane mirror, similarly to the case of using the concave surface; and the rate of change is almost the same. The characteristics depicted in FIG. **23** and FIG. **24** indicate that use of the polygonal plane mirror in the light radiating device **1G** also enables solidification or cauterization of the biological tissue PL, similarly to the light radiating device with the reflection surface being a concave surface.

(101) Various reflection surfaces may be employed as the reflection surface **31C** of the light radiating device **1G** according to the present embodiment. For example, a concave reflection surface formed by connecting three planes the normal line directions of which are different from one another can be employed, as illustrated in FIG. **25**. Thus, the shape of the reflection surface **31C** can be further approximated to a curved surface, and therefore a degree of condensation of a light beam (light ray IL) with respect to the y-axis direction can be enhanced.

(102) Further, the shape of the reflection surface **31C** of the light radiating device **1G** according to the present embodiment may be formed convex with respect to the y-axis direction by connecting three planes the normal line directions of which are different from one another, as illustrated in FIG. **26**. Thus, the shape of the reflection surface **31C** is not limited to concave or convex. For example, the shape of the reflection surface **31C** may be a parabolic surface.

(103) Further, the section of the reflection surface **31C** when the reflection surface **31C** is cut by the y-z plane orthogonal to the optical axis AX1 of the optical waveguide **20E** (x-axis direction) may be a concave polygonal line, as illustrated in FIG. **27**. Thus, a light ray IL in the x-axis direction as well as the y-axis direction can be reflected off the reflection surface **31C**, and a light

ray IL traveling in a z-axis direction can be condensed. Further, the section of the reflection surface **31C** may be a convex polygonal line. In this case, three planes the normal line directions of which are different may be connected or the polygonal line may be concave, in the x-axis direction.

#### Embodiment 8

(104) Next, Embodiment 8 of the present disclosure will be described. As illustrated in FIG. **28**, a light radiating device **1H** according to the present embodiment includes a light source **10D**, an optical waveguide **20E**, and a catoptric system **30C**. The light radiating device **1H** further includes a first optical condensing system **5**, a filter **6**, and a second optical condensing system **7**. In FIG. **28**, FIG. **29**, FIG. **30A**, FIG. **30B**, FIG. **31**, and FIG. **32**, the light source **10D** and the optical waveguide **20E** are depicted as a section including an optical axis **AX1** and being parallel to an x-z plane.

(105) The light source **10D** is a light guide including a plurality of optical fibers. The first optical condensing system **5** condenses a light beam **BM** emitted from the light source **10D** and causes the light beam **BM** to enter at one end of the optical waveguide **20E**. As illustrated in FIG. **29**, the light beam **BM** emitted from the light source **10D** becomes diffused light. The first optical condensing system **5** condenses the diffusing light beam **BM** and converts the light beam **BM** to a light beam **BM** closer to parallel light. The number of reflections of a light ray **IL** included in the light beam **BM** in the optical waveguide **20E** can be thereby reduced, and a decline in intensity of the light ray **IL** can be suppressed.

(106) The filter **6** eliminates light in the visible region from the light beam **BM** emitted from the first optical condensing system **5**. For example, eyes of an operator in an operating room, or the like can be thereby protected as much as possible.

(107) As illustrated in FIG. **30A**, the second optical condensing system **7** condenses the light beam **BM** emitted from the other end of the optical waveguide **20E** and causes the light beam **BM** to enter the catoptric system **30C**. In this case, a distance **L1** between the catoptric system **30C** and the second optical condensing system **7** is set to be greater than a focal distance **f1** of the second optical condensing system **7**.

(108) In this case, the light beam **BM** entering a reflection surface **31C** of the catoptric system **30C** becomes diffused light or condensed light. Therefore, the light beam **BM** reflected off the catoptric system **30C** is closer to a parallel light beam compared with the light radiating device **1A** (see FIG. **3**) not being provided with the second optical condensing system **7**. The above means that intensity of the light beam **BM** is uniform over a wide range with respect to a direction of an optical axis **AX2**. Extending, in a direction with respect to the optical axis **AX2** direction, the range in which intensity of the light beam **BM** is uniform can facilitate adjustment of a biological tissue **PL** to the range.

(109) When the sectional shape of the concave reflection surface **31C** when the reflection surface **31C** is cut by an incidence plane of a light ray **IL** entering along the optical axis **AX1** of the optical waveguide **20E** is approximated to a parabola, the reflection surface **31C** is placed in such a way that a focus position **C** of the second optical condensing system **7** matches a focus **f2** of the parabola, and a tilt angle of the reflection surface **31C** with respect to the optical axis **AX1** is set to 45°, light rays **IL** reflected off the reflection surface **31C** become parallel light parallel to the optical axis **AX2**. When the tilt angle of the optical axis **AX1** with respect to the reflection surface **31C** is not set to 45°, light reflected off the reflection surface **31C** does not become parallel to the optical axis **AX2** but can be brought close to parallel light along the optical axis **AX2**.

(110) Further, when the catoptric system **30C** is substituted with a catoptric system **30D** formed by connecting three plane mirrors **30D1**, **30D2**, and **30D3**, as illustrated in FIG. **30B**, light rays **IL** reflected off the catoptric system **30D** do not strictly become parallel light but can be brought close to parallel light along the optical axis **AX2**.

(111) The first optical condensing system **5**, the filter **6**, and the second optical condensing system **7** may be provided inside the optical waveguide **20E**, as illustrated in FIG. **31**. However, when the

effective diameter of the light beam BM emitted from the light source **10D** is large, the entire light beam BM emitted from the light source **10D** can be guided into the optical waveguide **20E** by providing the first optical condensing system **5** and the like outside the optical waveguide **20E** and setting the effective diameter of the first optical condensing system **5** larger than the effective diameter of the light beam BM.

(112) Further, while the light beam BM includes light at various wavelengths including an infrared radiation, optical paths of light rays IL in the first optical condensing system **5** and the second optical condensing system **7** vary with wavelength of the light. Therefore, out of light rays IL, only light at a specific useful wavelength such as an infrared ray is reflected off the catoptric system **30C** and becomes parallel light. Consequently, attenuation of light at a specific useful wavelength can be kept to a minimum.

(113) Furthermore, the catoptric system **30C** may be attachable and detachable. By removing the catoptric system **30C** as illustrated in FIG. **32**, a light ray IL can be condensed on the optical axis AX1 of the optical waveguide **20E**. Placing a biological tissue PL at the condensation position allows solidification or cauterization of the biological tissue PL.

(114) FIG. **23** and FIG. **24** respectively illustrate changes in a cauterization depth with respect to cauterization time and changes in a cauterization volume with respect to cauterization time in a case of removing the catoptric system **30C**. As illustrated in FIG. **23**, the changes in a cauterization depth with respect to cauterization time change parabolically. The rate of change is greater compared with the case of using the catoptric system **30C**. Further, as illustrated in FIG. **24**, the changes in a cauterization volume with respect to cauterization time also change parabolically. The rate of change is greater compared with the case of using the catoptric system **30C**.

(115) The size of the reflection surface **31C** of the catoptric system **30C** when the reflection surface **31C** is viewed from the optical axis AX1 direction of the optical waveguide **20E** is greater than the size of the section of the light beam BM emitted from the optical waveguide **20E**, as illustrated in FIG. **33**. Thus, the light beam BM emitted from the optical waveguide **20E** can be entirely reflected off the catoptric system **30C**. The above is normally achieved by causing the size of the reflection surface **31C** of the catoptric system **30C** when the reflection surface **31C** is viewed from the optical axis AX1 direction to be greater than the size of the section of the optical waveguide **20E** when the optical waveguide **20E** and the catoptric system **30C** are placed in proximity to each other. When the light beam BM emitted from the optical waveguide **20E** is a parallel light beam, the size of the reflection surface **31C** of the catoptric system **30C** has only to be equal to or greater than the size of the section of the optical waveguide **20E** orthogonal to the optical axis AX1.

(116) The light source **10D**, the optical waveguide **20E** or the catoptric system **30C** according to the present embodiment may be replaced by that used in another embodiment, or when the light beam of the light source **10D** is close to parallel light, the first optical condensing system **5** may be eliminated, and only the second optical condensing system **7** may be included.

(117) As described in detail above, the catoptric system **30A** reflecting and condensing the light beam BM sent to the other end of the optical waveguide **20A**, or the like is included in each of the aforementioned embodiments. Thus, a position where light is condensed by the catoptric system **30A** is intensely heated, and solidification or cauterization of a biological tissue PL can be performed at the position where light is condensed. Consequently, solidification or cauterization of the biological tissue PL at a focus F or in a region AR can be performed in a state of keeping the light radiating device **1A** or the like apart from the biological tissue PL without the light radiating device **1A** or the like being in tight contact with the biological tissue PL.

(118) According to each of the aforementioned embodiments, the sectional shape of the light beam BM can be converted into point-like (focus F), linear (region AR), or the like according to the shape of the reflection surface **31A** of the catoptric system **30A** or the like. Thus, the sectional shape of the light beam BM can be adjusted to the shape of a tissue to be solidified or cauterized, and therefore light can be radiated only to a biological tissue PL to be treated, and efficient

treatment can be provided.

(119) According to each of the aforementioned embodiments, light in the infrared region having high transmissivity for a biological tissue PL of a human or an animal is used, and therefore solidification or cauterization can be performed down to a deep spot in the biological tissue PL.

(120) According to each of the aforementioned embodiments, an affected part can be treated in a state of the device being kept apart from the affected part, and therefore a high degree of skill and experience, such as adjustment of a pressing force on the affected part, is not required when the light radiating device **1A** or the like is used.

(121) While a tubular member or a quartz rod is used as the optical waveguide according to each of the aforementioned embodiments, the present disclosure is not limited thereto. An optical fiber or a bundle of optical fibers may be used as the optical waveguide.

(122) Further, while the sectional shape of the optical waveguide in a prismatic shape is a hexagon, according to the aforementioned embodiment, the present disclosure is not limited thereto. The shape may be a triangle, a rectangle, a pentagon, or a polygon with seven angles or more.

(123) While the sectional shape of each of the reflection surfaces **31A** and **31B** of the catoptric systems **30A** and **30B** with respect to the optical axis **AX1** direction of the optical waveguides **20A** and **20B** is set to the elliptical arc **32**, according to each of the aforementioned embodiments, the present disclosure is not limited thereto. The shape has only to be a concave shape allowing the light beam **BM** to be condensed in the optical axis **AX1** direction.

(124) Further, while an average tilt angle of each of the reflection surfaces **31A** and **31B** of the catoptric systems **30A** and **30B** with respect to the optical axis **AX1** is set to about 45 degrees, according to each of the aforementioned embodiments, the present disclosure is not limited thereto. The average tilt angle may be other than 45 degrees.

(125) While the light beam **BM** emitted from each of the light sources **10A** and **10B** is set to light in the infrared region, according to each of the aforementioned embodiments, the present disclosure is not limited thereto. The light beam **BM** may be light in a wavelength region other than the infrared region. Further, the light beam **BM** may include light at wavelengths in the infrared region and another band.

(126) Further, each of the light radiating device **1A** and the like is assumed to be an optical scalpel performing solidification or cauterization of a biological tissue PL, according to each of the aforementioned embodiments, the present disclosure is not limited thereto. A radiation target may not be a biological tissue PL and may be another material, whether organic or inorganic.

(127) Further, while a surface formed by depositing a metal material such as aluminum is used as the reflection surface **21A** of each of the optical waveguides **20A** to **20E** or the like, according to each of the aforementioned embodiments, the present disclosure is not limited thereto. A metal material film surface formed by adhering a metal material by sputtering may be used as the reflection surface **21A**. The reflection surface **21A** has only to be a metal material film surface formed of a metal material, and a manufacturing method thereof is not limited.

(128) Furthermore, a reflection control film **22A** made up of silicon dioxide ( $\text{SiO}_2$ ) may be formed on the reflection surface **21A**, as illustrated in FIG. **34A**. Silicon dioxide has a relatively high surface reflectance and also has relatively high transmissivity for infrared light, and therefore the reflection control film **22A** can suppress attenuation of the light beam **BM** due to reflection off the reflection surface **21A**.

(129) Further, the thickness of the reflection control film **22A** may be set to  $\frac{1}{4}$  of a wavelength  $\lambda$  of a part of light rays **IL** included in the light beam **BM** emitted from the light source **10A** or **10B**, that is,  $\lambda/4$ , as illustrated in FIG. **34B**. For example, the part of light rays is a light ray at an unnecessary wavelength not contributing to heat generation in a radiation target PL. In this case, reflection intensity of a light ray **IL** at a specific wavelength  $\lambda$  (such as a blue light ray or a green light ray) out of light rays **IL** radiated to the radiation target PL can be reduced by an action described below.

(130) Specifically, part of a light ray **IL** at a wavelength  $\lambda$  included in the light beam **BM** emitted

from the light source **10A** or **10B** passes through the reflection control film **22A** as a light ray **IL'** and is reflected off the reflection surface **21A**; and the remainder is reflected off the surface of the reflection control film **22A** as the light ray **IL**. The light ray **IL'** passing through the reflection control film **22A** has a phase opposite to that of the light ray **IL** reflected off the surface of the reflection control film **22A** and is combined with the light ray **IL** reflected off the surface of the reflection control film **22A**; and the two rays cancel each other out. Consequently, intensity of the light ray **IL** at the specific wavelength  $\lambda$  can be lowered compared with light rays **IL** at other wavelengths in the light beam **BM** radiated to the radiation target **PL**. Thus, providing the reflection control film **22A** made up of  $\text{SiO}_2$  allows the optical waveguides **20A** to **20E** to have a filtering function of suppressing intensity of a light ray **IL** at a specific wavelength  $\lambda$ .

(131) The foregoing describes some example embodiments for explanatory purposes. Although the foregoing discussion has presented specific embodiments, persons skilled in the art will recognize that changes may be made in form and detail without departing from the broader spirit and scope of the invention. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. This detailed description, therefore, is not to be taken in a limiting sense, and the scope of the invention is defined only by the included claims, along with the full range of equivalents to which such claims are entitled.

(132) This application claims the benefit of Japanese Patent Application No. 2018-137030, filed on Jul. 20, 2018, the entire disclosure of which is incorporated by reference herein.

#### INDUSTRIAL APPLICABILITY

(133) The present disclosure is applicable to solidification or cauterization of a biological tissue or the like being a radiation target.

#### REFERENCE SIGNS LIST

(134) **1A, 1B, 1C, 1D, 1E, 1F, 1G, 1H** Light radiating device (optical scalpel) **5** First optical condensing system **6** Filter **7** Second optical condensing system **10A, 10B, 10C, 10D** Light source **20A, 20B, 20C, 20D, 20E** Optical waveguide **21A** Reflection surface **22A** Reflection control film **30A, 30B, 30C, 30D** Catoptric system **30C1, 30C2, 30D1, 30D2, 30D3** Plane mirror **31A, 31B, 31C** Reflection surface **31C1, 31C2** Reflection surface **32** Elliptical arc **33, 34** Circular arc **40** Optical fiber **50** Heat insulating part **51** Exterior cover **52** Spacer **53** Projection **54** Through hole **60** Operation device **61** Grip **62** Switch **65** Shutter **80** Optical fiber **AR** Region **AX1, AX2** Optical axis **BM** Light beam **CL** Curve **DL** Connecting line **F** Focus **IL, IL'** Light ray **IP1** Incidence plane **IP2** Plane **P** Incidence position **PL** Biological tissue (radiation target) **P1** First position **P2** Second position

## Claims

1. A light radiating device performing solidification or cauterization of a radiation target by radiating a light beam, the light radiating device comprising: a light source emitting the light beam; an optical waveguide including a tubular hollow member having an inner circumferential side wall extending between a first end and a second end along a first optical axis and provided with a first reflection surface including a metal material film formed on the inner circumferential side wall totally reflecting the light beam, causing the light beam emitted from the light source to enter a part enclosed by the inner circumferential side wall from the first end, and sending the light beam to the second end along the first optical axis; and a catoptric system reflecting the light beam sent to the second end of the optical waveguide and condensing the light beam on the radiation target; wherein the catoptric system includes a second reflection surface having two different sectional shapes including: a first sectional shape defined along a first axis and including a concave elliptical arc defined by an ellipse having a major axis defined along the first optical axis and configured to condense light rays entering each point on the concave elliptical arc onto a focus of the concave elliptical arc defined along a second optical axis orthogonal to the first optical axis; and a second



sectional shape defined along a second axis perpendicular to the first axis and including a concave circular arc including a radius such that light rays entering each point on the concave circular arc are condensed onto the focus of the concave elliptical arc defined along the second optical axis; wherein the optical waveguide is configured to reflect a light ray along the first optical axis at a first reflected position along the inner circumferential side wall and at a second reflected position along the inner circumferential side wall after being reflected at the first reflected position at a maximum emission angle of 30 degrees; and wherein the light radiating device includes a length defined between the light source and an incidence position defined on the second reflection surface of the catoptric system of a light ray entering along the first optical axis of the optical waveguide that is equal to four times a distance defined between the first reflected position and the second reflected position.

2. The light radiating device according to claim 1, wherein the second reflection surface of the catoptric system includes a metal film deposited on an enclosure.

3. The light radiating device according to claim 2, wherein the enclosure is formed of metal or resin.

4. The light radiating device according to claim 2, wherein the metal film is aluminum.

5. The light radiating device according to claim 1, wherein a film made up of silicon dioxide is formed on a surface of the metal material film.

6. The light radiating device according to claim 5, wherein a thickness of the film made up of silicon dioxide is  $\frac{1}{4}$  of a wavelength of a part of one or more light rays included in the light beam emitted from the light source.

7. The light radiating device according to claim 1, wherein the inner circumferential side wall is formed in a prismatic shape in the optical waveguide.

8. The light radiating device according to claim 1, wherein a heat insulating part is provided on an outer periphery of the optical waveguide and the catoptric system.

9. The light radiating device according to claim 8, wherein the heat insulating part includes: an exterior cover enclosing an outer periphery of the optical waveguide and the catoptric system; and a spacer inserted between the optical waveguide and the catoptric system, and the exterior cover in order to provide a gap between the optical waveguide and the catoptric system, and the exterior cover.

10. The light radiating device according to claim 9, wherein, on the spacer, a plurality of projections is provided on at least one of a part facing the optical waveguide and the catoptric system, and a part facing the exterior cover.

11. The light radiating device according to claim 1, wherein a size of the second reflection surface of the catoptric system when the second reflection surface is viewed from a direction of the first optical axis of the optical waveguide is equal to or greater than a size of a section of the light beam entering the catoptric system from the optical waveguide.

12. The light radiating device according to claim 1, wherein the light beam emitted from the light source includes an infrared radiation.

13. The light radiating device according to claim 12, wherein the light source is one of a halogen lamp, a xenon lamp, an infrared heater, and an infrared laser.

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